RAJALAKSHMI ENGINEERING COLLEGE

Kanish S 251801126-2025-AIDS K2

Started on	Sunday, 16 August 2026, 8:46 PM
State	Finished
Completed on	Sunday, 16 August 2026, 8:47 PM
Time taken	1 min 1 sec
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>

void function(int n)
{
    int c = 0;

    c++;   // i = n/2

    for(int i = n/2; i < n; i++)
    {
        c++;   // i < n

        for(int j = 1; j < n; j = 2 * j)
        {
            c++;   // j < n

            for(int k = 1; k < n; k = k * 2)
            {
                c++;   // k < n
                c++;   // counter statement
            }

            c++;   // j < n final check
        }

        c++;   // i < n final check
    }

    c++;       // final outer condition
    printf("%d", c);
}

int main()
{
    int n;
    scanf("%d", &n);

    function(n);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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